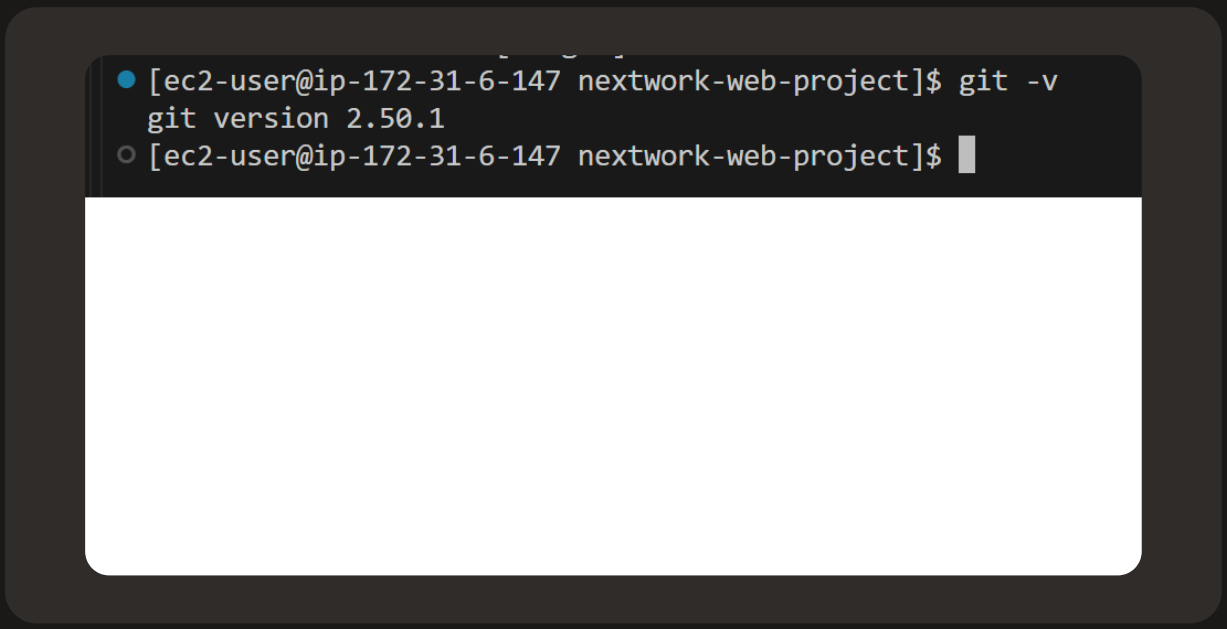
CONNECT A GITHUB REPO WITH AWS

By Maryna Maksimuk | October 2026



Introducing Today's Project!

Today, we are going to set up and use Git and GitHub to version-control my web application and prepare it for the CI/CD pipeline.

Key tools and concepts

I learned key AWS services such as EC2 along with concepts such as Git/GitHub, version control, CI/CD, cloud development, and automated application deployment.

Project reflection

This project took me approximately 2 hours. The most challenging part was setting up Git and GitHub authentication and understanding how the different tools work together. It was most rewarding to successfully make changes to my Java web application, commit them with Git, and push them to my GitHub repository.

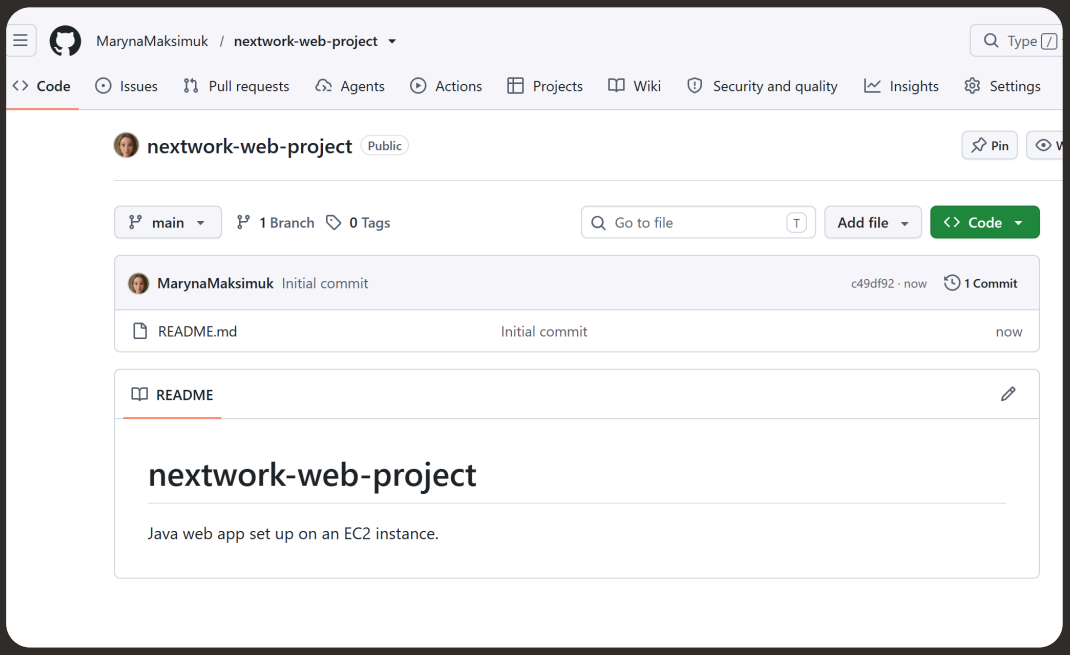
I did this project today to build a strong foundation in DevOps, Git/GitHub, AWS, and CI/CD by creating and managing a Java web application in the cloud.

I will start the next project tomorrow to continue building my DevOps and CI/CD skills.

Git and GitHub

Git is a version control system that tracks changes to my code and allows me to manage and collaborate on my project. I installed it on my EC2 instance using the command sudo dnf install git -y.

GitHub is a platform for storing and managing source code using Git version control. We are using GitHub to store our application code, track changes, collaborate on the project, and provide the source code that will later be used by our CI/CD pipeline for automated build, testing, and deployment.



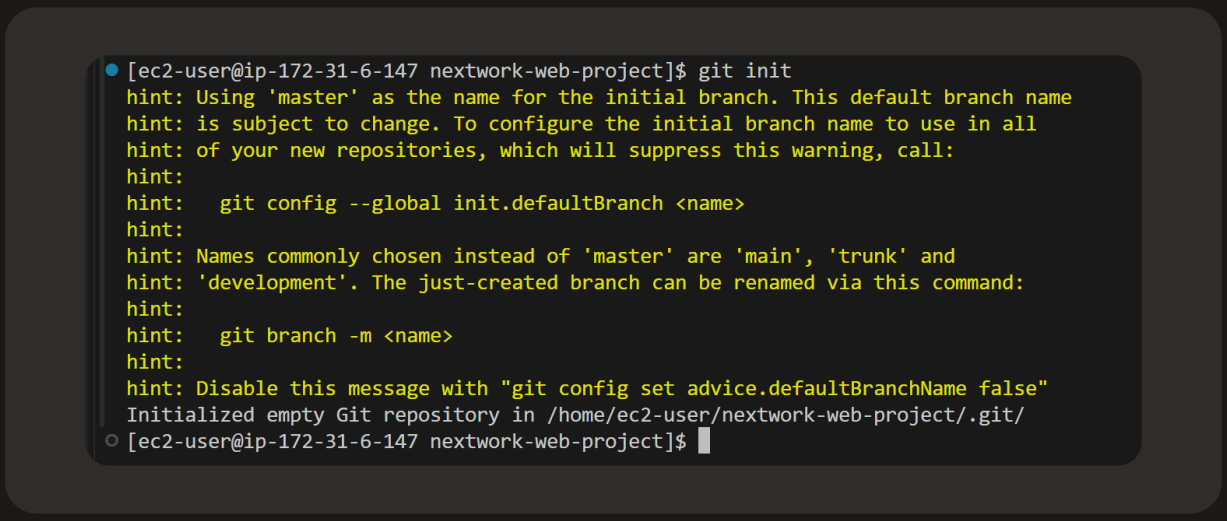
My local repository

A Git repository is a storage location that keeps your project files and tracks changes made to those files over time. We use a Git repository to manage the application's source code, keep a history of changes, and make it easy to collaborate and integrate the project with our CI/CD pipeline.

git init initializes a new Git repository in your current project folder. It creates a hidden .git directory that Git uses to track changes, maintain version history, and manage your project.

In our project, we use git init inside the web application folder so Git can start tracking our application code and prepare it to be connected to GitHub.

A Git branch is a separate line of development that allows you to make changes to your code without affecting the main version of the project. Branches are commonly used to develop new features, fix bugs, and test changes before merging them into the main (or master) branch.



To push local changes to GitHub, I ran three commands

git add

The first command I ran was git add . adds all changed and new files in the current directory to Git's staging area, preparing them to be committed.

git commit

The second command I ran was git commit -m "message" saves the staged changes to Git history, and -m means "message", which lets you add a description of what you changed.

git push

The third command I ran was git push -u origin master uploads your local master branch to the GitHub repository, and -u sets origin/master as the default upstream branch so future git push and git pull commands can be used without specifying the repository and branch.

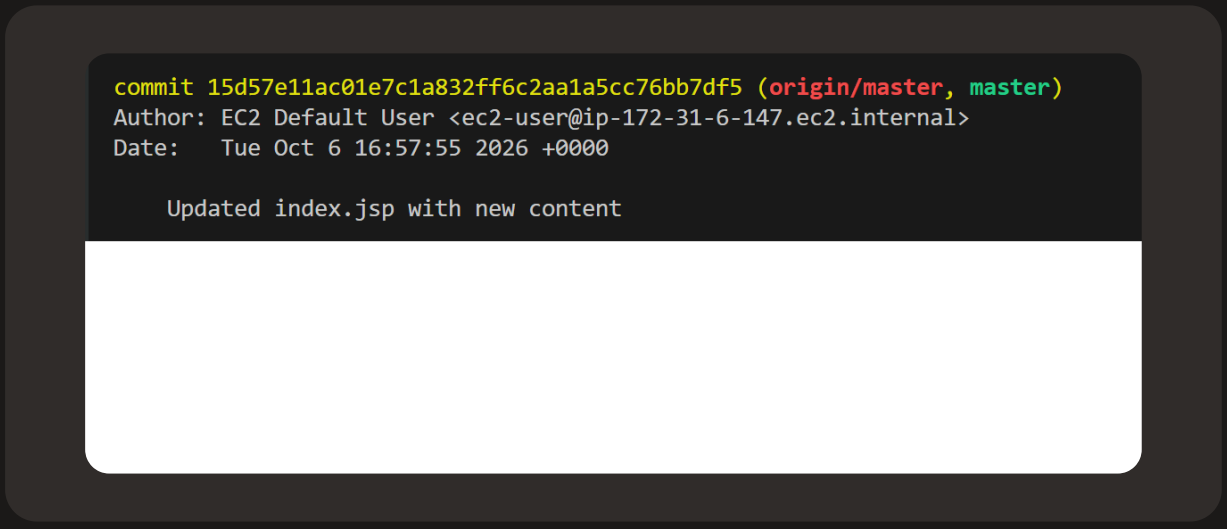
Authentication

Git asked for my username and password because the GitHub repository was initially configured to use HTTPS authentication, which requires credentials to verify that I have permission to push changes to the repository.

Local Git identity

Git needs a name and email to identify who made each commit and record that information in the Git history.

Running git log showed me the history of my commits, including the commit author, commit message, date, and unique commit ID; “EC2 Default User” appeared as the commit author because that is the Git name configured on my EC2 instance.

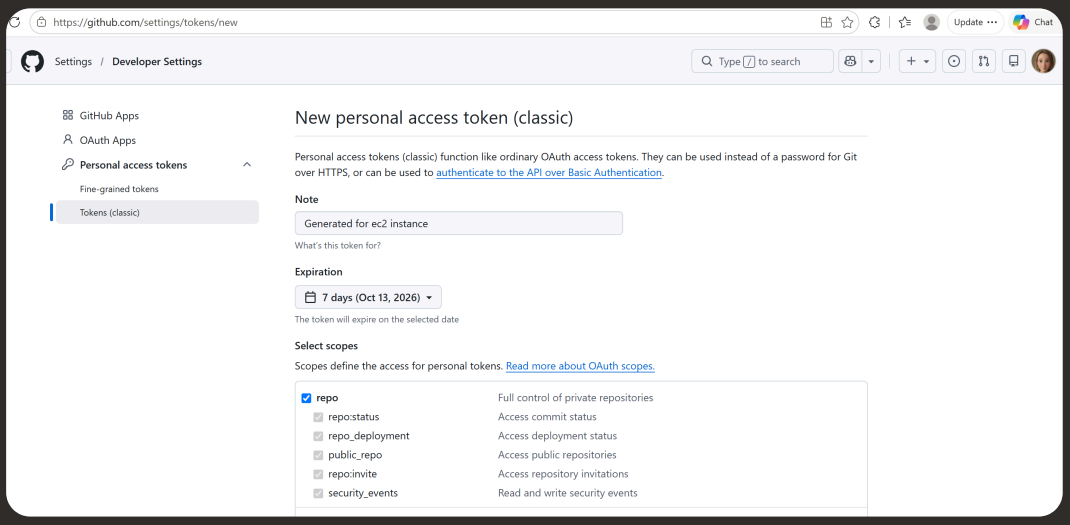


GitHub tokens

Authentication failed because GitHub no longer supports using a regular account password for Git operations over HTTPS; GitHub requires a personal access token (PAT) or SSH authentication instead.

A GitHub token is a secure authentication code that replaces your GitHub password when accessing a repository, and we use it to securely authenticate and allow Git to push changes to the GitHub repository.

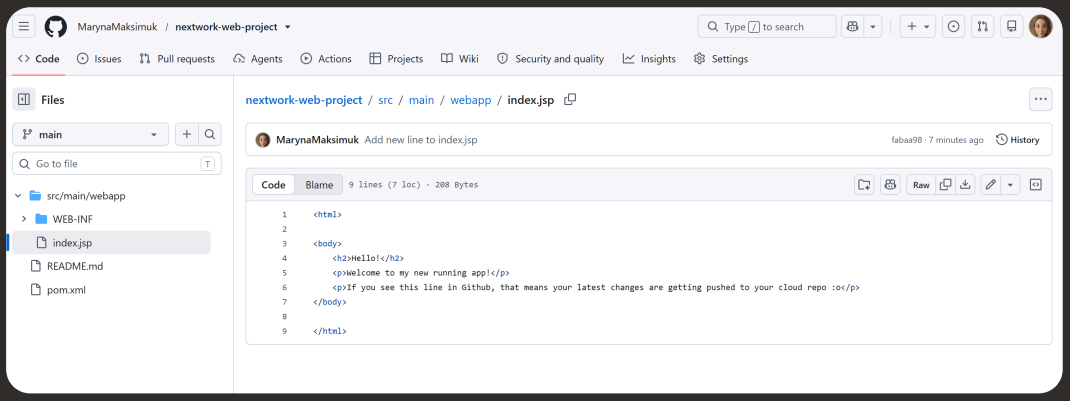
I set up a GitHub token by going to GitHub Settings → Developer settings → Personal access tokens → Fine-grained tokens, creating a new token with access to my repository and Read and write permission for repository contents.



Making changes again

I updated the index.jsp file in the nextwork-web-project by making a small change to the web application.

I had to commit my changes and push them to GitHub using git add, git commit, and git push so the updated index.jsp appeared in my GitHub repository.



Setting up a README file

A README file is a documentation file that explains what a project is, how it works, and how to use or set it up.

A README file is usually written in Markdown, a simple formatting language that makes it easy to create headings, lists, links, and other documentation.

My README.md contains six main sections: Introduction, Technologies, Setup, Contact, Conclusion, and Table of Contents. These sections explain the project, tools used, setup instructions, contact information, project goals, and provide navigation links.

